

SCENARIO FOUNDATIONS

Scenario Thinking

What **is** a climate scenario?

A climate scenario is a **plausible description of how the future may develop, based on a coherent and internally consistent set of assumptions** — not a prediction of what will happen, but a structured account of what could.¹

NGFS uses the same core scenario definition as the IPCC in its scenario glossary.² Both institutions treat scenarios as tools for exploring plausible futures under coherent assumption sets — not as competing forecasts of a single most-likely outcome.

Projection vs. Prediction

IPCC distinguishes a **climate projection** — a simulated response to a scenario of future emissions or forcing — from a **climate prediction**, which asserts the current state of the system evolves into a specific future state. Scenarios generate projections, not predictions.¹

Pathway vs. Scenario

A **pathway** is the trajectory of a specific variable over time (e.g. an emissions or concentration trajectory). A **scenario** is the broader narrative construct — assumptions about policy, technology and economics — that a pathway sits inside.³

EXPLORATORY VS. NORMATIVE

IEA distinguishes **exploratory scenarios** (“what happens if” — tracking real-world policy signals) from **normative scenarios** (“what must happen to reach a defined target”). Its Current Policies and Stated Policies scenarios are exploratory; Net Zero and its universal-access scenario are normative.⁴

DEEP UNCERTAINTY

RAND defines deep uncertainty as conditions where parties to a decision do not know, or cannot agree on: the system model, the probability distributions over inputs, or the value function used to rank outcomes.⁵ Climate futures sit inside this category — not ordinary, quantifiable risk.

NOT A FORECAST

A scenario is explicitly **neither a prediction nor a forecast**. It is a tool for understanding how a coherent set of conditions could unfold — developed fully on Page 2.

DOSSIER ARCHITECTURE · FOUR PILLARS

Dossier 1
Matter
Financial materiality

Dossier 2
Impact
Double materiality

Dossier 3
Future
Scenarios & transition risk

Dossier 4
Execution
Transformation & KPIs

¹ IPCC AR6 WG1, Annex VII Glossary; IPCC AR6 SYR Annex I Glossary.

² NGFS Scenarios Portal Glossary, ngfs.net.

³ NGFS Technical Documentation V5.0; IEA WEO-2025.

⁴ IEA, World Energy Outlook 2025, Scenarios.

⁵ RAND, Decision Making under Deep Uncertainty (DMDU) Program; RB-9701.

Scenarios Are Not Forecasts

Why is one future not enough?

A forecast asks: what is most likely to happen? A scenario set asks: what range of coherent futures should we understand the business against? These are different questions, and collapsing them into one produces false confidence.

Reducing a scenario exercise to a single “best case / base case / worst case” triad is a common but analytically weak simplification. It implies the three cases sit on one axis of severity, when in fact plausible futures differ across **multiple, only loosely correlated dimensions** — transition speed, physical-hazard trajectory, technology cost curves, regional policy divergence.

NGFS's own Phase V long-term scenario set spans seven distinct scenarios precisely because a single central case, or even three ordered cases, cannot represent the genuine width of plausible transition and physical-risk combinations.¹

DIMENSIONS THAT DIFFER, NOT ONE SEVERITY AXIS

TRANSITION SPEED

Early & gradual vs. late & abrupt policy tightening.

PHYSICAL TRAJECTORY

Hazard severity accumulating under a given emissions path.

TECHNOLOGY COST CURVES

How fast substitute technologies become competitive.

REGIONAL DIVERGENCE

Whether policy ambition is coordinated or fragmented across jurisdictions.

BEST / BASE / WORST — REJECTED

This framing implies scenarios rank on one severity axis. Climate futures do not: an orderly-but-slow transition and a disorderly-but-short one are not simply “better” or “worse” versions of each other — they carry different risk *types*, not just different risk *amounts*.

WHAT SCENARIOS ARE FOR

Not to predict which future occurs. To understand how the business would behave *if* a given coherent set of conditions materialized — and to make decisions that hold up across more than one such future.

KEY DISTINCTION

A scenario-based result is conditional on the scenario. It is not the company’s most likely future — a distinction this dossier returns to at every quantitative step, through Page 20.

¹ NGFS, Climate Scenarios for Central Banks and Supervisors – Phase V, Nov 2024.

METHOD, PRECISELY

Scenario Analysis, Sensitivity, Stress Testing, Reverse Stress Testing

Four related but **distinct** techniques.

SENSITIVITY ANALYSIS

Varies one input, holding others constant. No central bank or standard-setter reviewed (BIS, ECB, PRA, Fed) publishes a standalone formal definition — functional meaning is stable across usage even where formal sourcing is thin. EVIDENCE GAP

SCENARIO ANALYSIS

Varies a coherent, internally consistent set of assumptions together — “a tool to enhance critical strategic thinking,” exploring alternatives to business-as-usual.¹

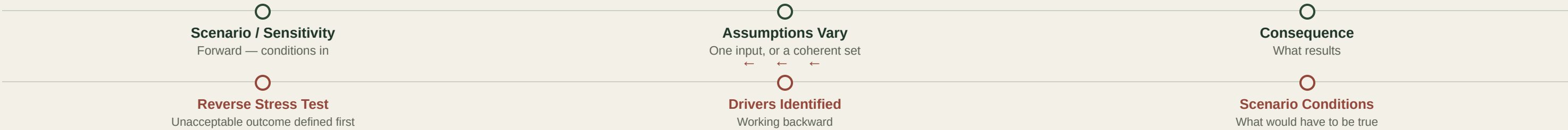
STRESS TESTING

Evaluates performance under a “severe but plausible scenario”¹ — severity is written into the definition itself, not merely contextual. The climate-specific boundary with scenario analysis remains contested.¹

REVERSE STRESS TEST

Starts from an unacceptable outcome and works backward to the conditions that would cause it — rather than starting from a scenario and projecting forward.²

DIRECTION OF REASONING



SEVERITY: DEFINITIONAL, NOT CONTEXTUAL

Severity is written into the stress-testing definition itself — “severe but plausible” — not an optional add-on. BIS itself notes the climate-specific boundary is contested: some regulators, including the Bank of England, describe severe exercises as “scenario analysis” rather than “stress tests.”¹

REVERSE STRESS TESTING, PRECISELY

PRA SS4/25 frames it around scenarios that would cause a firm’s business model to become *unviable* — a broader, more calibrated concept than simple insolvency, developed fully on Page 17.²

¹ BIS FSI Insights No. 34, footnote 7, p.4; BIS FSI, Climate Risks: Scenario Analysis.

² Bank of England PRA, Supervisory Statement SS4/25, paras 4.1–4.2.

THE NATURE OF THE PROBLEM

Uncertainty & Deep Uncertainty

Not all uncertainty is the same kind.

Ordinary risk assumes a known model, known probability distributions, and an agreed way to rank outcomes. Climate futures typically satisfy none of these conditions.

RAND's Robust Decision Making program defines **deep uncertainty** as conditions where the parties to a decision do not know, or cannot agree on: (1) the system model that relates actions to consequences, (2) the probability distributions over the model's inputs, or (3) how to value the consequences of different outcomes.¹

Climate scenarios sit inside this category on all three counts: the climate-economy system is contested, probabilities across NGFS's seven scenarios are not assigned, and how a company should weigh a severe-but-slow physical outcome against an abrupt-but-shorter transition outcome is itself a value judgment, not a calculation.

RAND'S THREE CONDITIONS OF DEEP UNCERTAINTY

SYSTEM MODEL

Parties disagree on which model relates actions to consequences — the climate-economy system itself is contested.

PROBABILITY DISTRIBUTIONS

No agreed probabilities across NGFS's seven scenarios or comparable sets.

VALUE FUNCTION

How to weigh a severe-but-slow physical outcome against an abrupt transition outcome is a judgment, not a calculation.

WHY THIS MATTERS FOR D3

Techniques built for ordinary, quantifiable risk (a single probability-weighted expected value) are the wrong tool here. Deep uncertainty calls for methods that test performance across a *range* of futures rather than optimizing against one — the robustness concept developed on Page 16.

WHAT DEEP UNCERTAINTY IS NOT

Not simply “a lot of uncertainty.” It is a qualitatively different condition — disagreement about the model itself, not just about parameter values inside an agreed model.

CARRIED FORWARD

Every technique introduced later in this dossier — scenario comparison, stress testing, reverse stress testing, robustness, optionality — is a response to deep uncertainty, not an attempt to eliminate it.

¹ RAND, Decision Making under Deep Uncertainty; Research Brief RB-9701.

PHYSICAL CLIMATE RISK

Physical Risk Foundations

What turns a hazard into risk?



IPCC frames these as dynamically interacting inputs, not a rigid pipeline — compounding and cascading relationships are explicitly allowed for (Page 7).¹

PRIMARY SOURCE

Hazard

“The potential occurrence of a natural or human-induced physical event or trend that may cause... damage and loss to property, infrastructure, livelihoods...”¹

ACUTE VS. CHRONIC

Acute: heatwaves, floods, wildfires, storms — discrete, event-driven.³ **Chronic:** gradual shifts — precipitation change, weather variability, ocean acidification, rising sea levels and average temperatures.³ The distinction is real, not cosmetic — a persisting post-event effect does not convert an acute hazard into a chronic one.

Sensitivity

The degree to which a system is affected, adversely or beneficially, by climate-related stimuli — the first of two components inside the IPCC vulnerability definition above.²

Exposure

“The presence of people; livelihoods... infrastructure; or economic, social, or cultural assets in places... that could be adversely affected.”¹

Vulnerability

“The propensity or predisposition to be adversely affected... sensitivity or susceptibility to harm and lack of capacity to cope and adapt.”²

NOT AUTOMATIC LOSS

Hazard does not equal financial loss. Vulnerability — including adaptive capacity — modifies how much of a given hazard actually transmits into business consequence. Transmission mechanics are developed on Page 13.

Adaptive Capacity

The ability of a system to adjust, moderate potential damage, or cope with consequences — the second component; low adaptive capacity raises vulnerability even where sensitivity is unchanged.²

FOUNDATION FOR PAGES 6–20

Every later transmission chain in this dossier — physical (Page 13) and transition (Page 6, 14) — builds on this hazard/exposure/vulnerability architecture and its explicit acute/chronic distinction.

¹ IPCC AR6 WGII Annex II Glossary.

² IPCC AR6 Synthesis Report, Annex I Glossary.

³ Basel Committee on Banking Supervision, Climate-related risk drivers and their transmission channels (d517), Glossary p.iv, §2.2.

TRANSITION RISK

Transition Risk

How does moving to a lower-carbon economy create risk?

IFRS S2's Basis for Conclusions (para. BC19) identifies transition risk as associated with **policy, legal, technology and market** changes arising from efforts to move to a lower-carbon economy.¹
Reputational risk is consistently added as a fifth category in practitioner guidance.²

PRIMARY SOURCE

INSTITUTIONAL GUIDANCE

Policy is a driver. Transition risk is the risk arising from the transition process — the same policy can create large risk for one company and none for another, depending on that company's specific exposure.

NOT NECESSARILY NEGATIVE

IFRS S2 requires disclosure of climate-related risks *and opportunities* symmetrically.¹ This dossier focuses on risk, but the transition also creates opportunity.

TIMING MATTERS

Magnitude depends on speed, sequencing and coordination — a slow, telegraphed shift allows reallocation over time; an abrupt one compresses the same adjustment into a shorter window. Developed on Page 8, 11.



CATEGORY, NOT YET CONSEQUENCE

Identifying a driver category is the first step, not the last. The same policy, technology or market shift can be immaterial to one company and severe to another — the transmission mechanics that decide which is developed on Page 14.

¹ IFRS S2, Basis for Conclusions, para. BC19; IFRS S2 scope.

² KPMG, IFRS S2 First Impressions, §3.3.1.

WHERE RISKS MEET

Physical + Transition Interaction

*Why can these **not always** be analyzed in silos?*

Compound Events

“The combination of multiple drivers and/or hazards that contribute to... risk.”¹ [PRIMARY SOURCE](#)

Risk aggregation is **not simply additive** — this follows directly from the compound-events definition. A 2025 academic review of NGFS's conceptual underpinnings finds that current NGFS long-term scenarios **do not fully integrate** physical risk within the economic projections that shape the transition pathway itself — an acknowledged limitation, not a solved problem.² [ACADEMIC EVIDENCE](#)

ILLUSTRATION — NOT A DOCUMENTED CASE

ANALYTICAL ILLUSTRATION

Physical water scarcity at a supplier, combined with stricter water regulation and a technology shift toward water-intensive production, can compound operating exposure beyond what any one condition suggests.

SIMULTANEOUS INTERACTION

Two or more hazards or drivers present at the same time — e.g. an acute physical event coinciding with a policy or market shock, straining response capacity on both fronts at once.

NOT SIMPLY ADDITIVE

Compound risk means two or more hazards or drivers interacting — simultaneously or consecutively — to produce combined effects the individual components alone would not predict. Risk A + Risk B ≠ Total Risk.

Cascading Impacts

“An extreme hazard generates a sequence of secondary events... whereby the resulting impact is significantly larger than the initial impact.”¹ [PRIMARY SOURCE](#)

ONE DOCUMENTED LINK

NGFS's own scenario design shows transition speed shaping eventual physical-risk severity — slower transitions carry materially higher physical-risk outcomes by 2050 across its scenario set.³ [PRIMARY SOURCE](#)

SEQUENTIAL INTERACTION

One hazard or driver follows another closely enough that recovery capacity from the first is not restored before the second arrives — the cascading-impacts mechanism above.¹

¹ IPCC AR6 Synthesis Report, Annex I Glossary.

² SUERF, Assessing the “Tragedy of the Horizons”: Conceptual Underpinnings of the NGFS Scenarios, 2025.

³ NGFS, Climate Scenarios for Central Banks and Supervisors – Phase V, Nov 2024.

Transition Pathways

Why does the path matter, not just the endpoint?

Different scenarios can target similar long-run climate outcomes while following materially different transition trajectories — that is not the same as reaching an identical endpoint.

NGFS's own Phase V narratives make this precise: **Net Zero 2050** targets an ambitious, immediate-policy pathway toward roughly 1.5°C; **Below 2°C** targets a gradually-increasing-stringency pathway with “roughly a 67% chance of limiting warming to below 2°C” — a related but numerically distinct target. **Delayed Transition** aims at a similar range to Below 2°C, but via a late, abrupt policy response rather than a gradual one.¹

A late pathway generally accumulates greater physical risk en route, even where the nominal long-run target is comparable (Page 7).

ORDERLY — NOT “GOOD”

Describes *how* a transition proceeds (early, coordinated, gradual) — not a value judgment on desirability.

FRAGMENTED WORLD

NGFS Phase V's “too little too late” case — delayed *and* divergent policy across regions, producing both higher transition risk and higher physical risk.¹

PATH DEPENDENCY

Small early events can lock in through increasing-returns dynamics — David (1985); Arthur (1989).² Developed fully on Page 11.

Same / similar long-run climate outcome

↓ can be reached via ↓

Gradual, early pathway — lower physical-risk accumulation

— or —

Late, abrupt pathway — higher physical-risk accumulation, compressed adjustment

Why timing matters: capital is committed at different points on different pathways; technology adoption curves differ; policy shocks differ in magnitude on a late path; and supply chains adapt differently depending on time available — developed fully on Page 11.

¹ NGFS, Climate Scenarios for Central Banks and Supervisors – Phase V, Nov 2024.

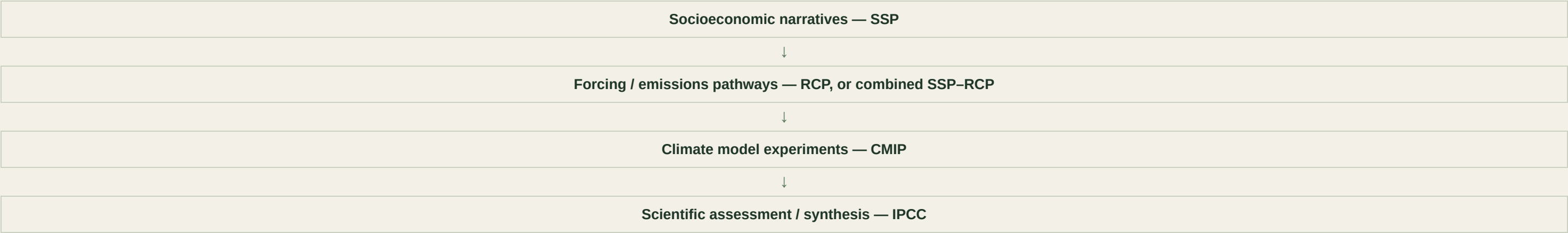
² David, P.A. (1985) AER 75(2); Arthur, W.B. (1989) Econ. J. 99(394).

THE ECOSYSTEM

Scenario Ecosystems

Different layers, different questions.

NGFS, IEA and the IPCC/SSP/RCP/CMIP substrate are not interchangeable — different producers, different purposes, different modelling architecture. An SSP is not simply “an IPCC scenario.”



ANALYTICAL SYNTHESIS

An analytical map of relationships — not a single universal pipeline. NGFS and IEA are independently modelled, parallel ecosystems, not downstream outputs of this chain.

LAYER	WHAT IT IS	TYPICAL OUTPUT	RELATIONSHIP
NGFS (Phase V)	Financial-sector climate-scenario framework	7 scenarios: transition + physical + macro-financial	Independent, parallel to IPCC substrate
IEA WEO-2025	Energy-system scenario framework	CPS, STEPS (exploratory); NZE, ACCESS (normative)	Independent; not SSP/RCP-based
SSP	Socioeconomic narrative/dataset	5 storylines (SSP1–5), population/GDP/urbanization	Paired with forcing target as SSPx-y
RCP	Radiative-forcing pathway	4 pathways, W/m² by 2100	Physical-outcome only; no narrative
CMIP	Model intercomparison framework	Coordinated model-run output	Runs SSP–RCP (or RCP-only) through models
IPCC	Scientific assessment / synthesis body	Assessment Reports	Assesses CMIP output; does not produce it

WHY THE DISTINCTION MATTERS IN PRACTICE

A document citing “RCP4.5” is describing a radiative-forcing pathway alone — no policy narrative, no macro-financial variables. A document citing “NGFS Below 2°C” is describing a coordinated transition-and-physical-risk scenario built for the financial sector. Treating one as interchangeable with the other conflates a physical-science input with a financial-sector output.

THE PRACTICAL CONSEQUENCE

Which layer a number comes from determines what it can be used for — translating a layer’s output into a company’s operating assumption is Page 12’s question, not this page’s.

Sources IIASA SSP Database; van Vuuren et al. (2011) Climatic Change 109(1); O'Neill et al. (2021) ESD 12; WCRP-CMIP; NGFS Phase V; IEA WEO-2025.

DEFINITION, PRECISELY

What Actually Defines a Scenario?

Not a collection of numbers.

A scenario is a coherent, internally consistent, purpose-built representation of how a set of conditions may evolve and interact — expressed through explicit assumptions and, typically but not universally, quantified through models.

Internal consistency is what separates a scenario from an arbitrary bundle of numbers: policy assumptions must be compatible with technology assumptions, which must be compatible with energy-system and emissions assumptions.

Narrative explains what changes, why, and in what sequence — distinct from quantitative output. IEA's own exploratory/normative split shows quantification is a matter of degree, not a binary requirement.

DIFFERENTIATION

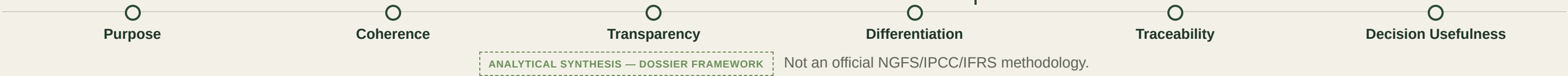
A scenario set should meaningfully challenge different key uncertainties — not maximize scenario count.

DECISION USEFULNESS

IFRS S2's own scenario-analysis requirement ties the choice of analytical approach to the entity's own circumstances — proportionality is built into the standard itself.¹

SCENARIO ≠ MODEL

NGFS's scenarios run through a *suite* of models (IAM + physical-risk + macroeconomic) — the scenario is the assumption package; the model is the computational tool.



SCENARIO ≠ COLLECTION OF NUMBERS

A scenario is not “carbon price = X, temperature = Y, GDP = Z.” It is a coherent representation of how a set of conditions may evolve and interact — internal consistency is precisely what separates the two.

¹ IFRS Foundation, Climate resilience and climate-related scenario analysis requirements in IFRS S2, March 2026.

DYNAMICS

Time Horizons, Path Dependency & Irreversibility

Why does **when** matter as much as **what**?

TIME HORIZONS

NGFS ties horizon choice to purpose: short-term (≤3–5yr) fits solvency assessment; long-term (10yr+) tests structural resilience — “the longer the scenario, the greater the uncertainty band.”¹

PRIMARY SOURCE

These are not the same thing and must not be conflated: physical irreversibility is a property of the climate system; economic irreversibility is a property of a company's capital decisions.

CENTURIES-SCALE PROCESSES

Ocean warming, ice-sheet loss and global sea-level rise are “irreversible for centuries to millennia” on human decision timescales — effectively fixed once triggered.²

PATH DEPENDENCY & LOCK-IN

The future development of a system is shaped by the path it has traced — David (1985), Arthur (1989).⁴ **Carbon lock-in**, per Unruh: industrial economies “locked into fossil fuel-based energy systems through... technological and institutional co-evolution driven by path-dependent increasing returns.”⁵ Not every asset automatically creates lock-in — it arises from co-evolving systems, not single investments.

IRREVERSIBILITY — PHYSICAL

“Many changes... are irreversible for centuries to millennia, especially... ocean, ice sheets and global sea level.”²

PRIMARY SOURCE

IRREVERSIBILITY — ECONOMIC

Capital commitments costly to reverse. Where uncertainty is present, irreversibility creates an **option value to waiting**.³

ACADEMIC EVIDENCE

FASTER-RESPONDING PROCESSES

Extreme-weather frequency and some regional precipitation patterns can shift on shorter timescales as forcing changes — physically distinct from the centuries-scale processes above, though both fall under “physical irreversibility.”²

TIPPING POINTS — PRECISELY

“Low-likelihood outcomes, such as ice-sheet collapse... cannot be ruled out and are part of risk assessment.”² Confidence varies by process — IPCC assigns “low confidence” to ice-sheet instability specifically, calling it “deep uncertainty.” Not every scenario framework captures these processes equally.

TIME HORIZON ≠ FORECAST HORIZON ONLY

A longer horizon does not simply push the same analysis further out — it changes uncertainty, interaction effects, path dependency, irreversibility, and decision flexibility all at once.

¹ NGFS, Guide to Climate Scenario Analysis – Update, Nov 2025, p.13, p.28.

² IPCC AR6 WGI SPM B.5, C.3; Technical Summary Box TS.1, TS.2, TS.9.

³ Arrow & Fisher (1974); Pindyck, investment-under-uncertainty literature.

⁴ David, P.A. (1985) AER 75(2), 332–337; Arthur, W.B. (1989) Econ. J. 99(394), 116–131.

⁵ Unruh, G.C. (2000) Energy Policy 28(12), 817–830.

THE BRIDGE

From Scenario to Operating Driver

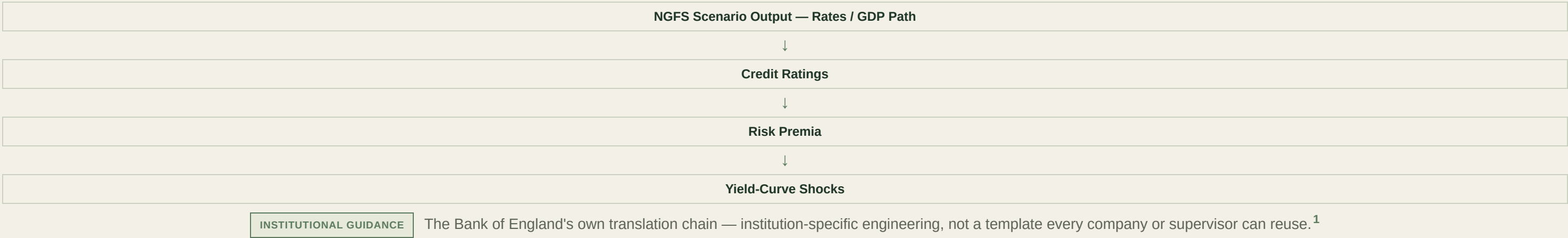
How do abstract variables become business assumptions?



Even NGFS's own current guidance stops short of an automatic translation: users must make “additional design choices to tailor the scenarios... complement NGFS outputs with internal data” — no end-to-end framework is provided.¹

PRIMARY SOURCE

ONE INSTITUTION'S TRANSLATION — NOT A TEMPLATE



REVENUE-SIDE

Volume · price · demand · mix · utilization

COST-SIDE

Energy · materials · carbon · logistics · insurance

ASSET-SIDE

Utilization · useful life · impairment · replacement

SUPPLY-CHAIN

Supplier availability · lead times · alternatives

SCENARIO TRANSLATION, EXPLICIT

ANALYTICAL ILLUSTRATION

Carbon price → company emissions exposure → applicable carbon cost → cost impact. A scenario's carbon-price output does not become a company’s cost line without this intermediate mapping.

COMPANY-SPECIFICITY

A global or sectoral scenario variable is necessary but not sufficient — it must combine with company-specific data (actual emissions, facility locations, supplier geography) to become an operating assumption. Reinforces Page 9.

THE MATERIALITY BOUNDARY

Page 12 asks through which mechanism a scenario touches the company — not whether the effect is financially material. That remains Dossier 1's question.

¹ NGFS, Guide to Climate Scenario Analysis – Update, Nov 2025, Box 1, p.10–11, p.23–24.

PHYSICAL TRANSMISSION

Physical Risk Transmission

How does a hazard reach the balance sheet?

ACUTE PHYSICAL RISK

Lethal heatwaves, floods, wildfires, storms — hurricanes, cyclones, typhoons, extreme precipitation.¹



A persisting post-event economic effect does not convert an acute hazard into a chronic one — a sustained sea-level-driven re-rating of flood exposure is the genuinely chronic mechanism.

DIRECT VS. INDIRECT

“Climate risks can directly impact... operations... [or] indirectly through exposures to counterparties and customers.”²

INSTITUTIONAL

SUPPLY-CHAIN PROPAGATION

Municipalities whose customer firms suffer drought show 1–2 p.p. lower GDP growth — even after controlling for local conditions. Moderate, not only extreme, shocks drive most of this transmission.³

ACADEMIC EVIDENCE

GEOGRAPHIC

Facilities or suppliers clustered in one exposed region share the same hazard footprint.

SUPPLIER

Few, non-substitutable suppliers transmit their own physical disruption directly into a company's operations.³

CUSTOMER

Revenue concentrated in customers or municipalities themselves exposed to the same hazard channel.

GEOGRAPHIC CONCENTRATION

Facilities, suppliers or infrastructure concentrated in one exposed region face greater correlated disruption — documented, for example, in persistent property-price declines in repeat-flood-vulnerable areas following Hurricane Sandy, an acute event with a lasting economic footprint.

¹ Basel Committee, Climate-related risk drivers and their transmission channels (d517), Glossary p.iv, §2.2.

² BIS FSI, Climate Risks in Banking: Transmission Channels, p.1–3.

³ Araujo, Linardi & Vissotto, BIS Working Paper 1260, Apr 2025.

TRANSITION TRANSMISSION

Transition Risk Transmission

How does a driver reach company economics?

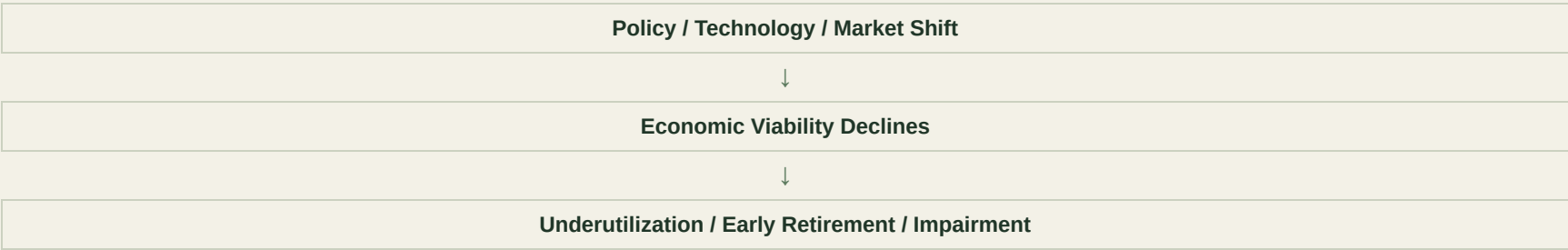


Each driver transmits via cost/constraint/incentive → operating response → economic consequence. Multiple drivers may interact, making their combined effect **different from** — not necessarily greater than — considering each in isolation. IPCC's compound-risk terminology (Page 7) does not establish a universal mathematical rule for transition channels specifically.

STRANDED-ASSET MECHANISM

“Assets that have suffered from unanticipated or premature write-downs, devaluations or conversion to liabilities.”¹

ACADEMIC EVIDENCE



TIMING

Speed, anticipation, sequencing and capital-cycle position all change the consequence of the same driver — directly connecting to Page 11's path-dependency findings.

BEFORE COMMITMENT

Driver arrives while capital is still uncommitted — assumptions can adjust before lock-in occurs.

AFTER LOCK-IN

Driver arrives once capital is committed — consequence is more severe; reversal is costly.

SHARED CHANNELS, ILLUSTRATIVE

COST

Carbon pricing and input-cost shifts reach the P&L directly — the mechanism developed quantitatively on Page 15.

COMPETITIVENESS

Technology substitution changes relative cost position versus peers, independent of any single policy action.

DEMAND

Market and reputational drivers shift customer preference — a revenue-side channel distinct from the cost-side ones.

CROSS-CHANNEL INTERACTION

Policy, technology and market drivers can interact through shared channels — cost, competitiveness, demand, investment, adoption, asset utilization — so analyzing any one driver alone can misstate a company’s actual combined exposure.

¹ Caldecott, Tilbury & Carey, Stranded Assets and Scenarios, Smith School Discussion Paper, Jan 2014.

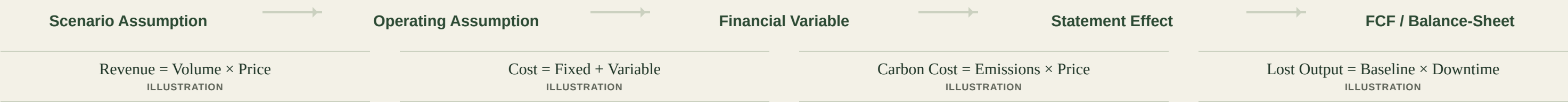
QUANTITATIVE BRIDGE

The Quantitative Transmission Bridge

How does a scenario become a number?

D1 FIREWALL

No DCF, WACC, terminal value, multiples, enterprise value or capital-structure mechanics. This page shows only where a scenario enters the financial model — not how the model is built.



Baseline vs. scenario-specific assumption: a company establishes a current baseline (e.g. emissions intensity), then applies scenario-specific assumptions to project how that baseline might shift — emissions-accounting methodology itself is not taught here.

These four relationships are illustrative only — which one applies depends on which operating driver from Page 12 is being translated (revenue-side, cost-side, asset-side or supply-chain), and no single equation covers every driver category.

UNCERTAINTY LAYERS

Scenario (which future) · **Model/structural** (which framework) · **Parameter** (values of estimated relationships) · **Company-specific** (translation to firm assumptions). These accumulate across layers and can interact to widen the range of outcomes — no specific mathematical aggregation rule is asserted.

ANALYTICAL SYNTHESIS

BALANCE-SHEET FRAMING

NGFS: static balance-sheet assumption suits short-term (3–5yr) exercises; dynamic assumption suits 10yr+ horizons, at the cost of added “behavioural uncertainty.”¹

PRIMARY SOURCE

CONDITIONAL, NOT A FORECAST

A scenario-based financial result is conditional on the scenario, the model, and the assumptions. It is not the company’s most likely future FCF — the strongest conclusion of this dossier's quantitative chapter, and the one Page 20 builds its closing statement on.

¹ NGFS, Guide to Climate Scenario Analysis – Update, Nov 2025, §3.4, p.28.

FROM SCENARIO TO TEST

Stress Testing & Scenario Comparison

How do we turn scenarios into structured tests?

PLAUSIBILITY ≠ SEVERITY

Stress testing's value comes from severity relative to a defined threshold — not from plausibility relative to a central forecast. A highly extreme scenario can be a valuable stress test precisely *because* it is severe, even where it is clearly not the central case. “Worst case = best stress test” is not the rule — an implausible extreme is uninformative; the most plausible scenario is rarely the most useful stress test.

COVERAGE, NOT COUNT

A well-designed set challenges different transition, physical, policy, technology and demand assumptions — NGFS's own four-family structure is evidence of coverage-by-design. Two scenarios producing nearly identical consequences add little decision value, whatever their narrative variety.

ROBUSTNESS ≠ OPTIMIZATION

RAND: rather than a “predict-then-act” search for the best outcome under one assumed future, Robust Decision Making identifies “strategies that perform well over a wide range of futures.”¹

ACADEMIC EVIDENCE

RAND's own water-management case reduced high-cost outcomes from 120 to 30 across a wide scenario range — not by finding the theoretical optimum, but by widening acceptable performance.¹



ANALYTICAL SYNTHESIS

A visualization simplification — NGFS's own seven-scenario set (Page 9) spans more dimensions than any 2×2 matrix.

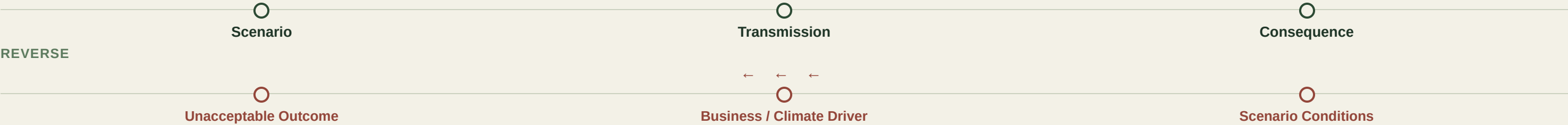
¹ RAND, Research Brief RB-9701, Robust Decision Making for Planning Under Deep Uncertainty.

WORKING BACKWARD

Reverse Stress Testing & Vulnerability Mapping

*Start from **failure**. Work backward.*

FORWARD



FAILURE ≠ BANKRUPTCY

PRA SS4/25 frames reverse stress testing around scenarios causing a business model to become *unviable* — broader than simple insolvency.¹ **PRIMARY SOURCE** Corporate thresholds — capital breach, liquidity threshold, covenant breach, service-level failure, production threshold — are an adaptation of this framework, not PRA's own list. **ANALYTICAL ILLUSTRATION**

CONCENTRATION

Geographic, supplier, customer, technology and infrastructure concentration raise susceptibility to correlated disruption — concentration does not automatically equal high risk; it becomes risk only once combined with an actual hazard or driver.

ILLUSTRATIVE FAILURE CONDITIONS

CAPITAL BREACH

LIQUIDITY THRESHOLD

COVENANT BREACH

SERVICE-LEVEL FAILURE

PRODUCTION THRESHOLD

WHAT REVERSE STRESS TESTING DOES NOT CLAIM

Identifying the conditions that would cause failure is not the same as estimating how likely those conditions are — this is a vulnerability-mapping exercise, not a probability exercise. Pairing it with the forward scenario set (Page 16) shows whether any plausible scenario approaches the identified failure conditions.

VULNERABILITY ≠ RISK

Vulnerability: “the propensity or predisposition to be adversely affected... lack of capacity to cope and adapt.”² **Risk:** the potential consequence, produced only once a hazard/driver interacts with exposure and vulnerability. A company can be highly vulnerable with low realized risk if no hazard is currently present.

THE USEFUL OUTPUT

Not another worst-case narrative — a specific answer to which conditions would have to combine for the business to cross the failure threshold.

¹ Bank of England PRA, SS4/25, paras 4.1–4.2.

² IPCC AR6 SYR, Annex I Glossary.

CHOOSING UNDER UNCERTAINTY

Strategic Optionality, Adaptation & Stranded Assets

What is flexibility worth?

CONCEPTUAL ONLY

Irreversibility + uncertainty → value of flexibility / waiting. No option-pricing mathematics, no Black–Scholes, no derivatives.

WAIT VS. ACT

Waiting preserves information; acting early can reduce future exposure, capture learning-curve advantage, avoid lock-in, and secure scarce resources. Neither is universally superior — the balance depends on irreversibility, the speed uncertainty resolves, and competitive dynamics.

STAGED INVESTMENT

ANALYTICAL SYNTHESIS

Small initial commitment → learning → reassess → scale / change / abandon.

ADAPTATION ≠ MITIGATION

Adaptation: “adjustment to actual or expected climate... to moderate harm or exploit... opportunities.” **Mitigation:** “intervention to reduce emissions or enhance... sinks.”¹ **PRIMARY SOURCE** Mitigation reduces future climate change itself; adaptation reduces harm from change already occurring.

ROBUST × FLEXIBLE

FLEXIBLE

FRAGILE Easy to change, poor if wrong future hits first	ROBUST & FLEXIBLE Generally preferred, often costlier
NEITHER Weak on both dimensions	ROBUST, INFLEXIBLE Overbuilt but hard to reconfigure

INFLEXIBLE

Optionality ≠ Optimization: optimization chooses the best strategy under assumed conditions; optionality preserves the ability to change course as uncertainty resolves.

Stranded assets — decision implications: the same eventual endpoint can arrive at very different speeds (Page 8) — a company committing under an “orderly” assumption faces different stranding risk than under a delayed-then-abrupt pathway. Staged investment and delayed commitment are design responses to this risk, not merely descriptions of it.

¹ IPCC AR6 SYR, Annex I Glossary.

Transition Pathways & Resilience Assessment

Is the business model resilient?

Resilience — IPCC

“The capacity of interconnected... systems to cope with a hazardous event... responding or reorganizing in ways that maintain their essential function, identity and structure.”¹

PRIMARY SOURCE

Climate Resilience — IFRS S2

“A company's capacity — strategically and operationally — to manage climate-related risks and benefit from... opportunities.”²

PRIMARY SOURCE

Robustness (Page 16) is a *pre-event* property evaluated across a scenario set; resilience explicitly includes the *post-event* dynamic of responding and reorganizing. Related, not identical.

Scenario Set		Exposures	Transmission	Outcomes	Thresholds	Resilience
DIMENSION		ORDERLY		DISORDERLY		HOT HOUSE WORLD
Revenue impact		—		—		—
Asset exposure		—		—		—
Strategic flexibility		—		—		—
ANALYTICAL ILLUSTRATION	Template structure — no actual values populated.					

ADAPTIVE CAPACITY

Resilience depends partly on flexibility, redundancy, access to capital, technology and supply-chain alternatives — directly the “capacity to cope and adapt” component of the Page 5 vulnerability definition. Higher adaptive capacity, lower vulnerability, for a given exposure.

NOT EVERY BUSINESS, EVERY SCENARIO

Which of NGFS’s seven scenarios matter depends on the company’s own exposure profile (Pages 12–14) — coverage, not scenario-count-maximization (Page 16).

RESILIENCE ≠ ABSENCE OF RISK

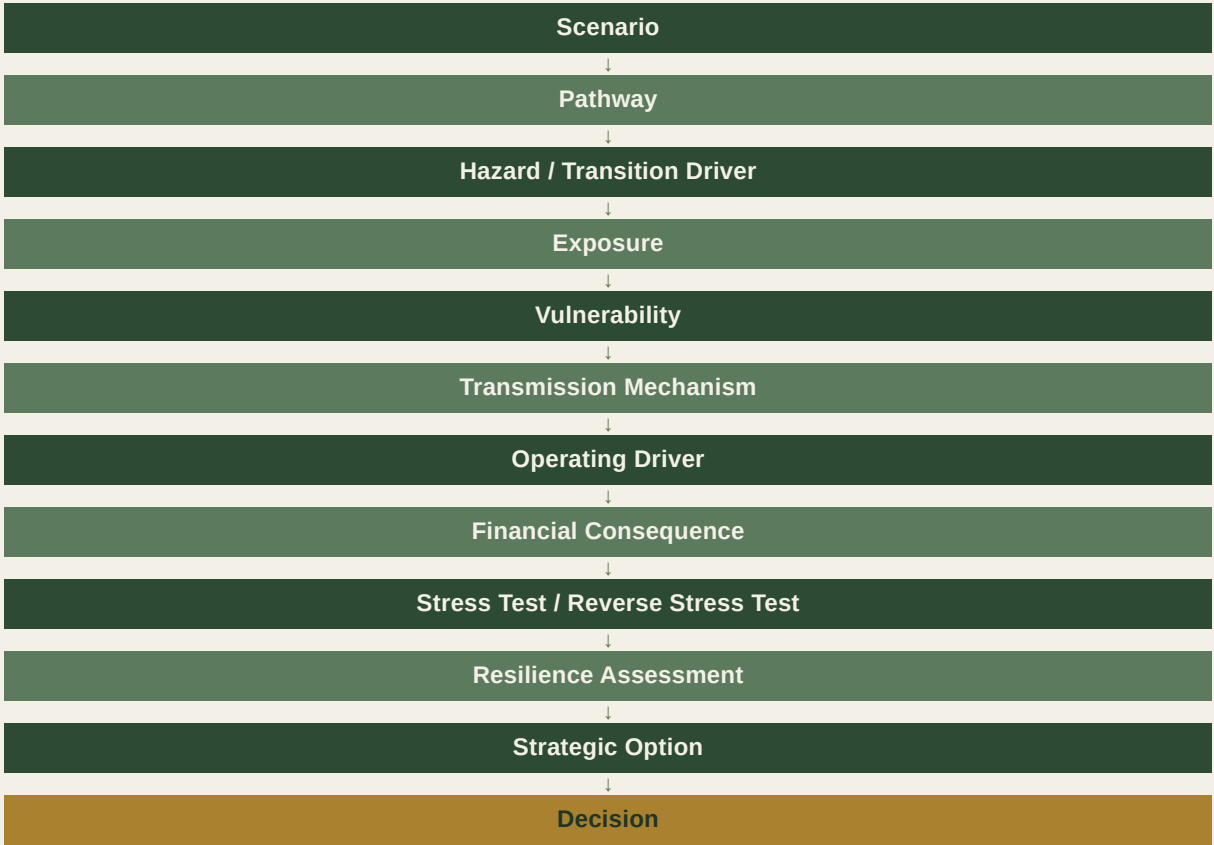
A resilient company may still experience losses, disruption, higher costs or impairment. Resilience means the business remains viable and can absorb and adapt within defined limits — not that the effects do not occur.

¹ IPCC AR6 SYR, Annex I Glossary.

² IFRS Foundation, Climate resilience factsheet, March 2026.

The Complete Climate Scenario Decision Framework

*How does it all fit **together**?*



ANALYTICAL SYNTHESIS — DOSSIER FRAMEWORK

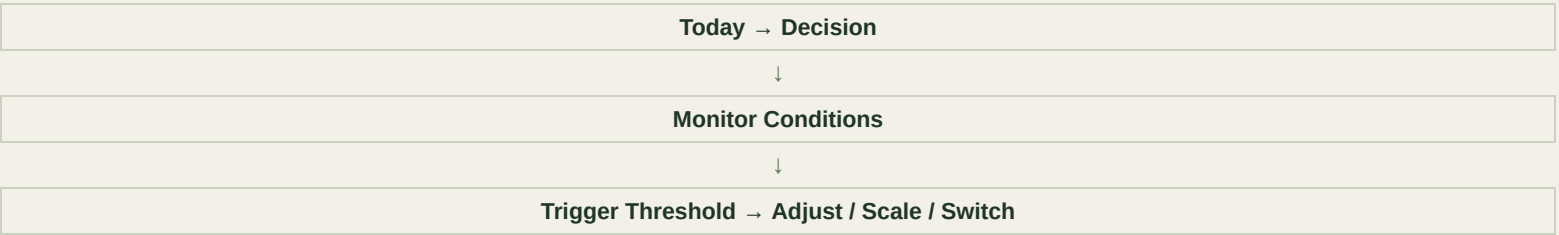
Not an official NGFS / IPCC / IFRS framework.

DECISION TYPES

No-regret — benefits even absent climate change. **Reversible / flexible** — keep the cost of being wrong low. **Safety-margin** — reduce vulnerability at null/low cost. **Soft** — institutional/financial, low inertia. **Shorter-horizon** — reduce exposure to long-run uncertainty.¹ [ACADEMIC EVIDENCE](#)

TRIGGER-BASED ADAPTATION

Dynamic Adaptive Policy Pathways: identify “adaptation tipping points” — conditions where a current strategy fails — and a monitoring plan of “signposts and triggers” for switching pathways as conditions unfold.² [ACADEMIC EVIDENCE](#)



THE OUTPUT

Not “which scenario will happen” — but which decisions remain defensible across the range of plausible futures.

The purpose of scenario analysis is not to predict the future. It is to understand how different futures could change the business — and to make decisions that remain defensible when the future does not unfold as expected.

¹ Hallegatte, S. (2009) Global Environmental Change 19(2), 240–247.

² Haasnoot, Kwakkel, Walker & ter Maat (2013) Global Environmental Change 23(2), 485–498.